

Registers copied by reference

<https://github.com/Qiskit/qiskit/issues/4177>

ROW 94

Registers copied by reference #4177

New issue



Closed

#4178



Cryoris opened on Apr 18, 2020 · edited by Cryoris

Edits ...

Information

- Qiskit Terra version: 0.13.0
- Python version: 3.7.5
- Operating system: MacOS Catalina

What is the current behavior?

The registers in the circuit are copied via reference. Modifying the registers in the copied circuit changes the original circuit.

Note that this causes Aqua algorithms such as the VQE to fail, if passed a quantum circuit and run on the QASM simulator. This happens because the Algorithm may add a classical register called 'c' to a copy of the Ansatz wavefunction multiple times, but since the original circuit is modified, the adding fails as a register called 'c' already exists.

Steps to reproduce the problem

```
circuit = QuantumCircuit(1)
circuit.x(0)
copied = circuit.copy()
copied.add_register(ClassicalRegister(1, name='c'))
print(circuit.cregs)
```



produces

```
[ClassicalRegister(1, 'c')]
```



What is the expected behavior?

Copy `qregs` and `cregs`.

Suggested solutions

```
# L1176 in quantumcircuit.py
cpy = copy.copy(self)
cpy.qregs = self.qregs.copy()
cpy.cregs = self.cregs.copy()
```



Cryoris added **bug** on Apr 18, 2020

Cryoris mentioned this on Apr 18, 2020

[↗ Properly copy registers on circuit copy #4178](#)

Cryoris added **priority: high** on Apr 19, 2020

Cryoris added this to the **0.14** milestone on Apr 19, 2020

lucian0 added this to To do in **Visualization and Jupyter Tools** on Apr 20, 2020

lucian0 added this to To do in **Circuits** on Apr 20, 2020

lucian0 removed this from To do in **Visualization and Jupyter Tools** on Apr 20, 2020

Assignees

No one assigned

Labels

bug **priority: high**

Type

No type

Projects

No projects

Milestone

0.14

Closed on Apr 30, 2020, 100% complete

Relationships

None yet

Development

[↗ Properly copy registers on circuit copy](#)
Qiskit/qiskit

Notifications

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Participants



Properly copy registers on circuit copy #4178

<> Code

Merged mergify merged 6 commits into Qiskit:master from Cryoris:4177-copy-registers on Apr 20, 2020

Conversation 3 Commits 6 Checks 0 Files changed 2 +17 -0



Cryoris commented on Apr 18, 2020

Contributor

Summary

Fixes #4177.

Details and comments

On QuantumCircuit.copy the registers were passed via reference and not copied. Hence changing the copied object led to changes in the original circuit.



fix copy

Unverified

9c5d8c5

Cryoris requested review from 1ucian0, ajavadia, ewinston, kdk and mtreinish as code owners 5 years ago

Cryoris added 2 commits 5 years ago

Merge branch 'master' into 4177-copy-registers

Verified

dca156a

Merge branch 'master' into 4177-copy-registers

Verified

f389173

Cryoris added bug priority: high labels on Apr 20, 2020

Cryoris added this to the 0.14 milestone on Apr 20, 2020

ajavadia previously approved these changes on Apr 20, 2020

View reviewed changes

ajavadia added the automerge label on Apr 20, 2020

Merge branch 'master' into 4177-copy-registers

Verified

a31f9ce

kdk requested changes on Apr 20, 2020

View reviewed changes

kdk left a comment

Member

Can you add a test covering this?



Reviewers

kdk



ajavadia



1ucian0



ewinston



mtreinish



Assignees

No one assigned

Labels

bug

Changelog: Bugfix

priority: high

Projects

None yet

Milestone

0.14

Development

Successfully merging this pull request may close these issues.

Registers copied by reference

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3 participants

